27 May 1959
SUPERSEDING
MIL-T-45127 (Ord)
26 June 1958

MILITARY SPECIFICATION

TIMER, INTERVAL TD-117/M

1. SCOPE

1.1 This specification covers one type of 5-minute delay timer, designated as Timer, Interval TD-117/M, hereinafter referred to as "delay timer".

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools, for Ordnance Materiel, Packaging of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-F-45133	Fire Control Components and Subassemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

Ordnance Corps

C7622606	Timer, Interval, TD-117/M schematic Diagram
D7617256	Timer, Interval TD-117/M

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Material shall be in accordance with the drawings and specifications forming a part of this specification.

3.2 Electrical.

3.2.1 Continuity.- The delay timer (Drawing D7617256) shall demonstrate continuity in accordance with electrical schematic, Drawing C7622606.

3.2.2 High potential.- There shall be no puncture of insulation sufficient to cause failure of the delay timer when the insulation between ungrounded circuits and ground, and between isolated circuits is subjected to a potential of 1000 volts plus twice the rated circuit voltage root mean square (r.m.s) at a frequency of 60 cycles per second (c.p.s.). Insulation resistance requirement shall be checked immediately after this requirement.

3.2.3 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (d.c.) applied.

3.3 Performance.- The delay timer, when set-up in accordance with pretest information (section 4), shall meet the following requirements. (All symbols referred to are those shown on Drawing C7622606.)

3.3.1 Start relay, K1.- Start relay, K1 shall operate at 85 volts or less.

3.3.2 Motor operation.

(a) Five minutes ± 15 seconds (delay time) after K1 has been activated, the circuit between terminals 9 and 11 of P1 shall be closed.

(b) One minute, 40 ± 2 seconds after K1 has been deactivated the motor time delay control shall be capable of providing a recurrent delay time of 5 minutes ± 15 seconds.

(c) The motor time delay control shall be capable of providing a recurrent delay of 2.4 ± 0.3 minutes after an interruption of 48 ± 2 seconds.

3.3.2.1 Reopening interval.- The switch contacts on the motor shall be capable of opening within 4 seconds after the 120 volt power is interrupted.

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3.4 Vibration.- The delay timer shall suffer no damage and shall be capable of meeting the performance requirements of this specification after vibration as follows:

Vibration Frequency* (cycles per sec.)	Total Excursion (inches)	Acceleration (g's)	Time (hrs. ea. direction)
2 - 10	0.060	---	1
11 - 20	0.030	---	2
21 - 400	-----	2.5	2

*Uniform rate of change shall not be greater than 1 cycle per 3 seconds nor less than 1 cycle per 5 seconds.

3.5 Workmanship.- Workmanship shall be in accordance with the applicable workmanship requirements of Specifications MIL-F-13926 and MIL-F-45133.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor inspection.- Unless otherwise specified herein, the contractor is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the ex amination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.1.2 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides:

(a) The quality assurance required by the applicable drawings and and specifications listed in the contract or order.
(b) Contractor's quality standards for acceptability.
(c) Methods for continually evaluating product quality.
(d) Means for continual evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

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4.1.3 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of surveillance of the operations to determine whether the practices, methods and procedures of the written inspection plan are being properly applied, and Government product inspection, when deemed necessary to measure quality product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Inspection provisions.- Quality assurance provisions specified in Specifications MIL-F-13926 and MIL-F-45133 shall apply where applicable and are not in conflict with the provisions specified herein.

4.2.1 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.2.2 Lot formation.- Unless otherwise specified by the contracting officer or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.2.3 Examination.- The classification of defects in Tables I and II shall constitute the minimum inspection requirements to be performed prior to acceptance.

4.2.3.1 All rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

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TABLE I - CLASSIFICATION OF DEFECTS

Normal Inspection - Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

Reduced Inspection - Use Inspection Level L8 and Sampling Plan Table IV-A of Standard MIL-STD-105

Major AQL 1.0	Req. Par.	Q.A. Par.	Test Method
1. Completeness of assembly	per dwg.		Visual
2. Continuity	3.2.1	4.4.1	SME
3. High potential	3.2.2	4.4.2	SME
4. Insulation resistance	3.2.3	4.4.3	SME
5. Start relay, K1	3.3.1	4.4.4.1	SME
6. Motor operation	3.3.2	4.4.4.2	SME
7. Reopening interval	3.3.2.1	4.4.4.3	SME
8. Workmanship	3.5	4.4.7	Visual

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L-8, and Sampling Plan Table V of Standard MIL-STD-105

Major AQL 1.5	Req. Par.	Q.A. Par.	Test Method
1. Vibration	3.4	4.4.5	SME

4.3 Inspection equipment.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor shall also maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment and to use it for Government verification inspection or acceptance inspection.

4.3.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of 10 percent of the specified tolerance. In cases where the 10 percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, a waiver or deviation from this requirement will be considered upon presentation to the contracting officer.

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4.4 Test methods and procedures

4.4.1 Continuity.- A continuity check of the delay timer circuits shall be made utilizing standard measuring equipment to determine conformance with the requirements specified in 3.2.1.

4.4.2 High potential.- The unit shall be tested in accordance with Method 301 of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.2.2.

4.4.3 Insulation.- Immediately after satisfactorily withstanding the high potential test, the insulation resistance test shall be performed on the unit under test in accordance with Method 302, Test Condition B, of Standard MIL-STD-202 to determine conformance with the requirements specified in 3.2.3.

4.4.4 To determine compliance with 3.3, all symbols referred to are those shown on Drawing C7622606 or on Figure 1.

4.4.4.1 Start relay, K1.- An ohmmeter, 0-500 ohms, plus or minus 5 percent shall be connected to Terminals 3 and 5 of P1 on the unit under test. With a gradual increase in voltage from a power supply, 120 volts ac variable, 60 to 120 volts manually, 0.2 ampere, 400 cps single phase, the relay K1 shall operate. The voltage as indicated on a Voltmeter, ac, 0-15 volts plus or minus 5 percent shall not exceed the voltage specified in 3.3.1. Relay operation shall be indicated by a zero reading (short) indication on the ohmmeter.

4.4.4.2 Motor operation.- The voltage from power supply used in 4.3.4.1 shall be set at 120 volts and a power supply, minus 28 volts, 0.4 ampere, within plus or minus 5 percent, shall be turned on. An ohmmeter, as specified in 4.3.4.1 shall be connected to 9 and 11 of P1 on the unit under test. The 120 volt switch, single pole single throw, shall be closed and the motor run until the ohmmeter shows a zero reading (short). This delay interval shall be in accordance with the delay time specified in 3.3.2(a).

4.4.4.2.1 The 120 volt power shall be interrupted by opening switch, single pole single throw, for one minute, 40 plus or minus 2 seconds as measured with a stop watch, with 1 and 1/10 second divisions. When the switch is closed immediately after the interruption, the motor delay control shall provide a five-minute plus or minus 15 second delay. The stop on the timer shall be adjusted so that for any interruption equal to or greater than one minute 30 seconds the delay time shall be as specified in 3.3.2(b).

4.4.4.2.2 The timing motor shall provide a recurrent delay of 2.4 plus or minus 0.3 minutes after an interruption of the time specified in 3.3.2(c).

4.4.4.3 Reopening interval.- The switch contacts on the motor shall open within the time specified in 3.3.2.1 after the 120 volt power is interrupted.

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4.4.5 Vibration.- The delay timer Drawing D7617256 shall be mounted using its own mounting provisions on a vibrating machine capable of producing a motion at a constant frequency conforming to the frequencies specified in 3.4 with an amplitude of 0.030 and 0.060 inch total excursion. The testing machine shall be equipped with a medium to assure that mounting is free from resonances over the test frequency range. The medium shall hold the unit under test in the normal operating position. The unit under test shall be vibrated successively along each of its three mutually perpendicular directions that are respectively parallel to the edges of the unit with an acceleration, duration, and frequency conforming to the conditions in 3.4. The measuring instruments of the testing equipment shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The applied frequency shall be changed at a uniform rate of change in compliance with the requirements of 3.4. Upon completion of the test, the unit under test shall be inspected to determine that the requirements specified in 3.4 are met.

4.4.6 Quality assurance for packaging shall be as specified in Specification MIL-P-14232.

4.4.7 Workmanship.- Workmanship shall comply with the requirements of 3.5.

5. PREPARATION FOR DELIVERY

5.1 The timer shall be cleaned, preserved, packaged, packed, and marked in accordance with Specification MIL-F-14619 P150. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The delay timer is intended to delay the application of high voltage to the tracking-modulator tubes for 5 minutes. This delay allows sufficient time for the modulator and magnetron tubes to heat.

6.1.1 Application.- The delay timer is a component of the following assemblies and systems:

	Assembly	System
(a)	Radar Set Group	Antiaircraft Fire Control Systems M33C (90mm) and M33D (120mm)
(b)	Power Supply Group OA-730/MPA-4	Radar Course Directing Central AN/MSW-1
(c)	Cabinet, Component Test (No. 6)	Sets, Kits, and Outfits for Ground Equipment

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing.
(c) Applicable stock number.

6.3 Definitions.

6.3.1 Standard measuring equipment (SME).- Standard measuring equipment (SME) includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

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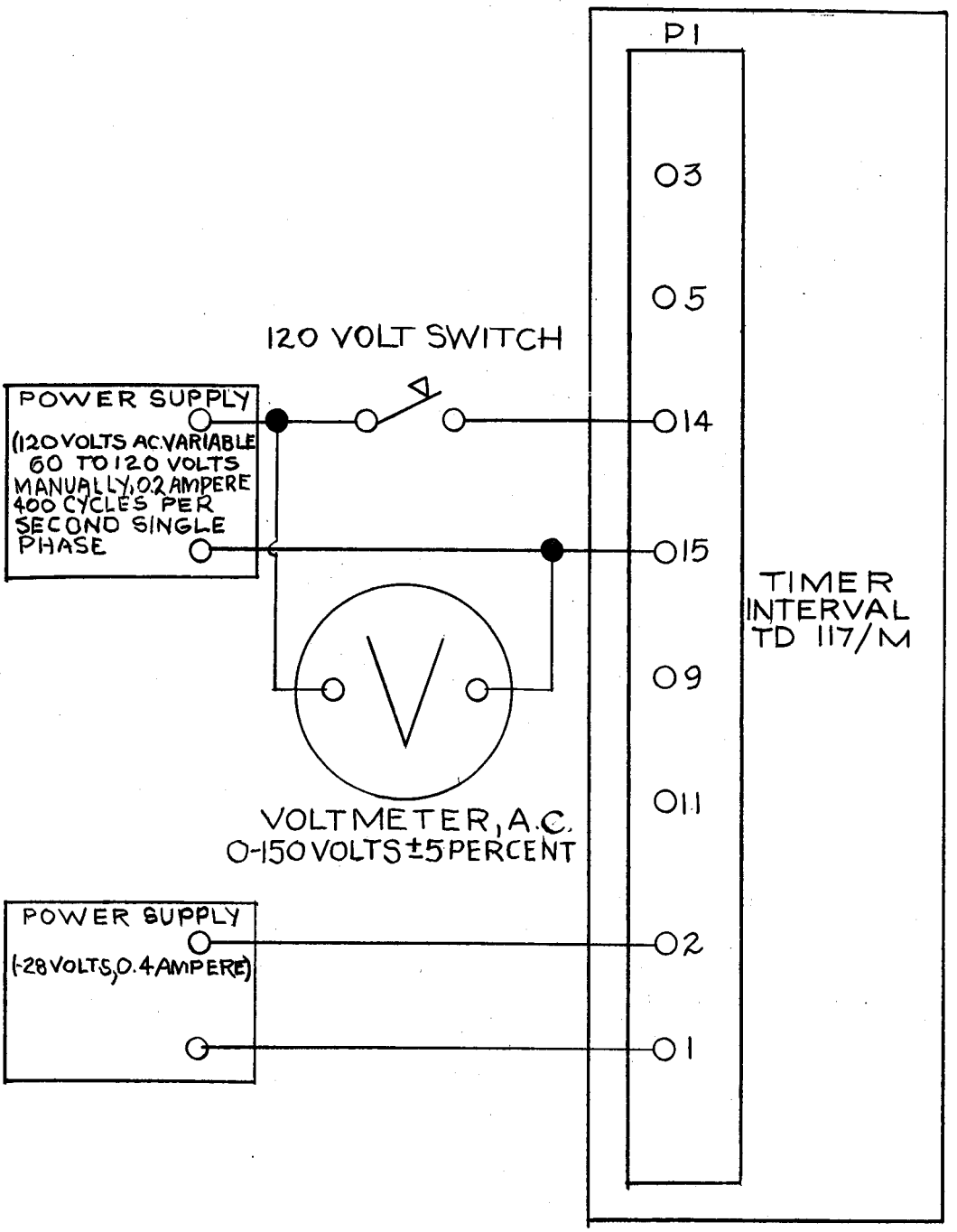


FIGURE 1

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